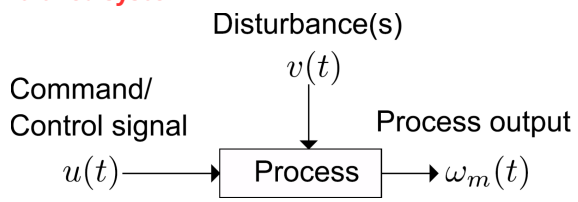
SW 1: Control Theroy and Automation

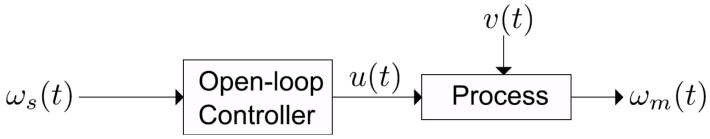
A control system regulates a process using sensors, controllers, actuators, and feedback.

Control theory designs controllers to regulate and optimize dynamic systems.

Uncontrolled system



Open-loop controlled system



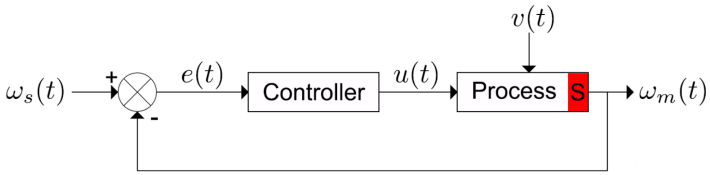
Advantages

- Simple structure: no sensors or controllers needed
- Low cost: fewer components and less complexity

Disadvantages

- No adaptation: cannot correct changes or disturbances
- Unstable performance: quality may decrease over time

Closed-loop controlled system



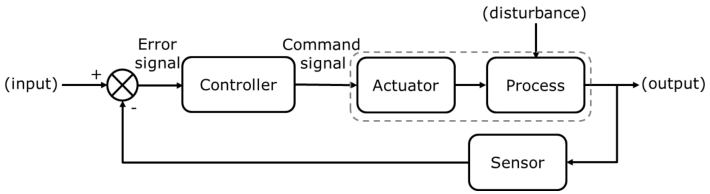
Advantages

- High accuracy: continuous correction improves precision, performance, and efficiency
- Adaptation: reacts to load changes, disturbances, or operating conditions

Disadvantages

- Complex structure: sensors, controllers, and feedback are required
- Higher cost: more components increase total cost
- More maintenance: components need calibration and maintenance

Extended block diagram



Components

Controller

Input: error $e(t)$
Output: command signal $u(t)$

Actuator

Input: command signal $u(t)$
Output: process input signal

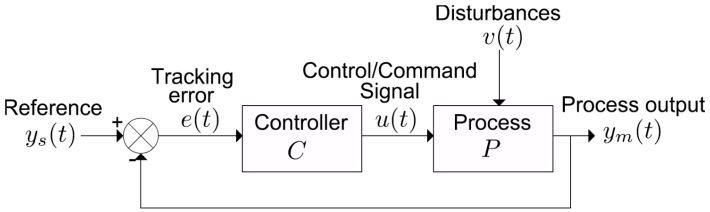
Process

Input: process input signal
Unwanted output: disturbance/noise $v(t)$
Output: actual process output $y_m(t)$

Sensor

Input: actual process output $y_m(t)$
Output: signal to be compared to the desired output signal

Feedback loop in a closed-loop control



$$e(t) = y_s(t) - y_m(t)$$
$$u(t) = C(e(t))$$
$$y_m(t) = P(u(t), v(t))$$

$y_s(t), e(t), u(t), y_m(t)$: Signals
P, C : Systems

Signals vs System

Signals

- Carry information through variations in a physical quantity
- Described as functions of one or more variables
- Can be natural or artificial and are used in communication, measurement, and control

System

- Interconnection of components used to perform a specific function
- Takes input signals, processes or transforms them, and produces output signals
- Describes the relationship between inputs and outputs

Automation pyramid

Level 0. Field/process level
Level 1. Control devices level
Level 2. Supervisory control level
Level 3. Manufacturing planning and execution system (MES)
Level 4. Management/Business level or ERP
Level 5. Digital transformation and AI-driven intelligent ecosystems (cloud level)

Control engineering

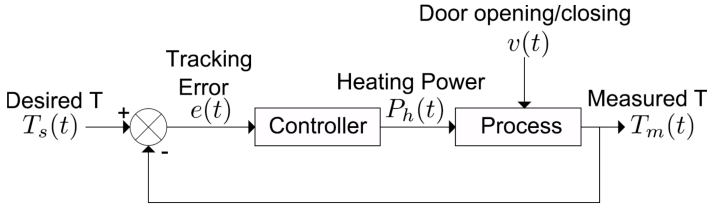
Levels 0-1: real-time control of physical processes using sensors, measurement systems, actuators, and controllers

Automation

Levels 2-5: integration, supervision, optimization, planning, and management of production and business processes

SW2: Good controllers

Sauna example, closed-loop control



Disturbance rejection

- Ability to reduce the effect of internal or external disturbances on the system output
- Keeps the output close to the desired reference

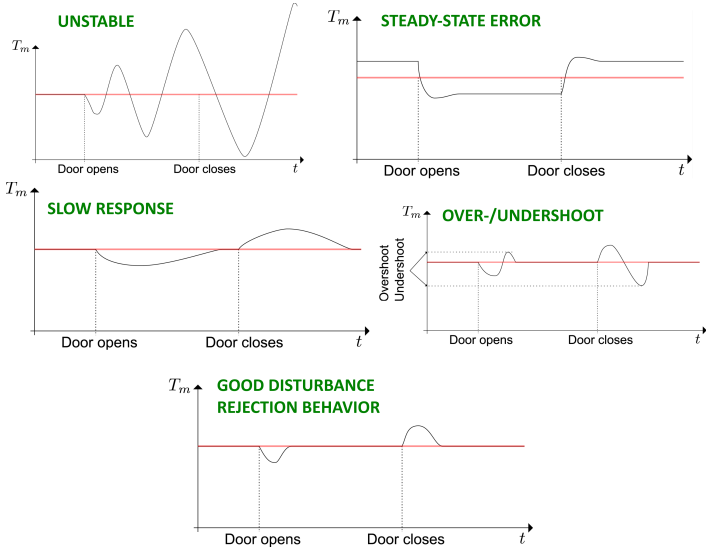
Characteristics

- Disturbance types: internal component changes or external environmental changes
- Effectiveness: controller detects and compensates disturbances
- Control strategies: feedback control reacts after disturbance; feedforward control acts before it affects the output

Performance metrics

- Stability: system remains robust under disturbances
- Steady-state error: final deviation from the reference is minimized
- Overshoot/undershoot: excessive output deviations are reduced
- Rise/settling time: response is fast and stable

Graphical representation



Reference tracking

- Ability to follow the desired reference signal accurately
- Tracking error: difference between reference and output

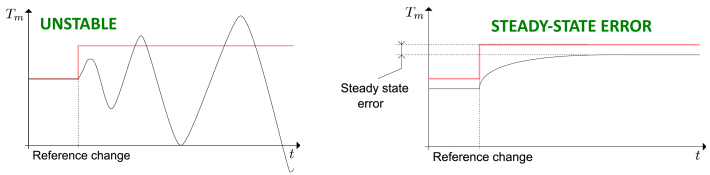
Characteristics

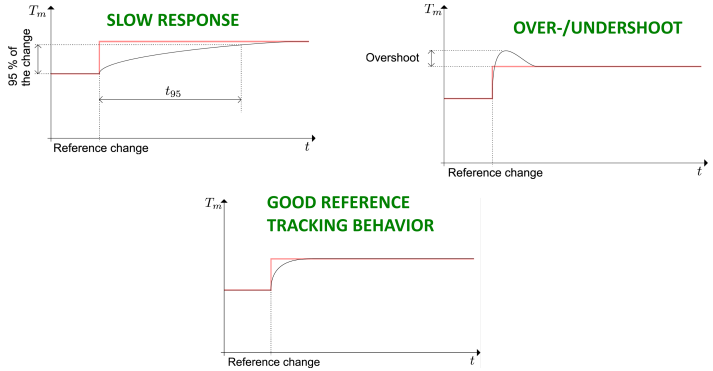
- Tracking accuracy: output follows the reference with minimal deviation
- Effectiveness: controller reduces tracking error under different conditions
- Control strategies: feedback corrects deviations; feedforward anticipates reference changes

Performance metrics

- Stability: tracking remains consistent and robust
- Steady-state error: long-term deviation from the reference is minimized
- Overshoot/undershoot: excessive deviations during tracking are reduced
- Rise/settling time: output reaches the reference quickly and stably

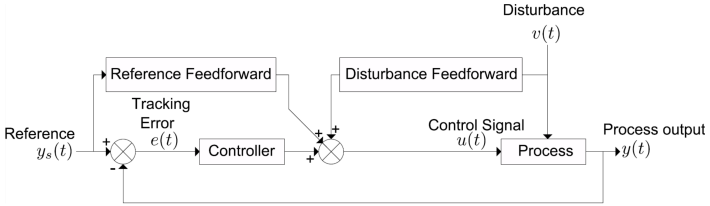
Graphical representation





Feedforward control

- Predictive control strategy that compensates known influences before they affect the output
- Improves performance by anticipating changes instead of only reacting to errors



Reference feedforward

- Compensates reference changes before they affect the system
- Improves tracking by adjusting the control input in advance

Disturbance feedforward

- Compensates measurable disturbances before they affect the output
- Uses a precomputed correction based on the expected disturbance effect

SW3: Process model development

Damped mass-spring example

Differential equation

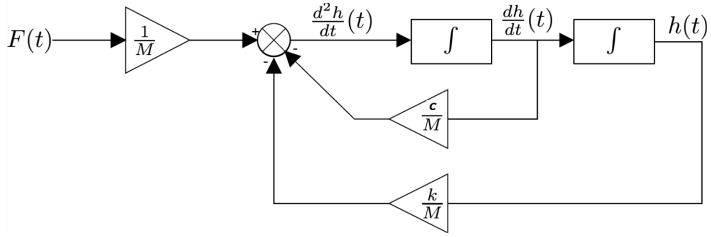
Step 1: Set the differential equation

$$M\frac{d^2h}{dt^2} + c\frac{dh}{dt} + kh = F$$

Step 2: Isolate the term with the highest differential order

$$\frac{d^2h}{dt^2} = \frac{F}{M} - \frac{kh}{M} - \frac{c}{M}\frac{dh}{dt}$$

ODE diagram



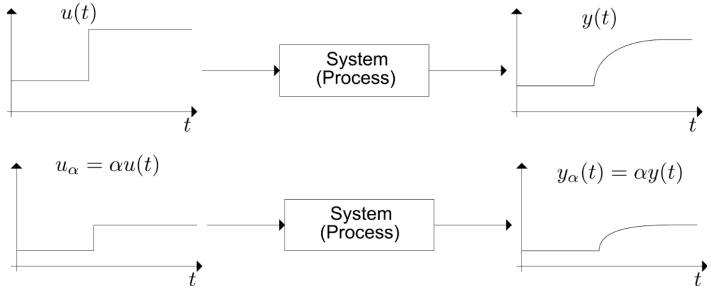
SW4: Signals and Systems

Homogeneity (scaling principle)

Scaling the input by a constant also scales the output by the same constant:

- If $u(t) \xrightarrow{h} y(t)$, then $\alpha u(t) \xrightarrow{h} \alpha y(t)$
- If $y(t) = h(u(t))$, then $h(\alpha u(t))$, that is, $h(\alpha u(t)) = \alpha h(u(t))$

Graphical visualization



Homogeneity proof examples

Homogeneous system

$$u(t) \xrightarrow{h} y(t) = 4u(t)$$
$$h(\alpha u(t)) = 4(\alpha u(t)) = \alpha(4u(t)) = \alpha h(u(t))$$
$$\alpha u(t) \xrightarrow{h} \alpha y(t) \qquad \square$$

Non-homogeneous system

$$u(t) \xrightarrow{h} y(t) = u^3(t)$$
$$h(\alpha u(t)) = (\alpha u(t))^3 = \alpha^3 y(t) \neq \alpha y(t)$$
$$h(\alpha u(t)) \neq \alpha h(u(t)) \qquad \square$$

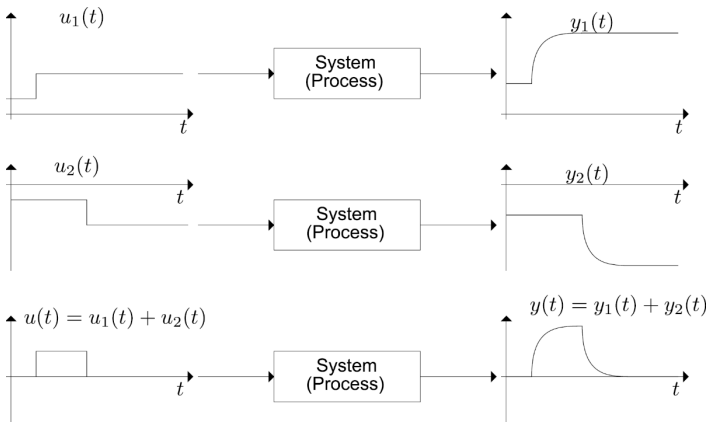
Additivity (superposition principle)

The output of summed inputs equals the sum of the individual outputs:

- If $u_1(t) \to y_1(t)$ and $u_2(t) \to y_2(t)$, then
$$u_1(t) + u_2(t) \to y_1(t) + y_2(t)$$
- If $y_1(t) = h(u_1(t))$ and $y_2(t) = h(u_2(t))$, then
$$h(u_1(t) + u_2(t)) = y_1(t) + y_2(t)$$

That is
$$h(u_1(t) + u_2(t)) = h(u_1(t)) + h(u_2(t))$$

Graphical visualization



Additivity proof examples

Additive system

$$u_1(t) \xrightarrow{h} y_1(t) = 2u_1(t)$$
$$u_2(t) \xrightarrow{h} y_2(t) = 2u_2(t)$$
$$h\big(u_1(t) + u_2(t)\big) = 2\big(u_1(t) + u_2(t)\big)$$
$$= 2u_1(t) + 2u_2(t) = y_1(t) + y_2(t)$$
$$u_1(t) + u_2(t) \xrightarrow{h} y_1(t) + y_2(t) \qquad \square$$

Non-additive system

$$u(t) \xrightarrow{h} y(t) = \sin(u(t))$$
$$h(u_+(t) + u_2(t)) = \sin(u_1(t) + u_2(t)) \neq \sin(u_1(t)) + \sin(u_2(t)) \qquad \square$$

Linear system

A system is linear if it satisfies both the homogeneity (scaling) and additivity principles.

Time-invariant system

System behavior does not change over time. A time-shifted input produces the same time-shifted output:

- If $u(t) \to y(t)$, then
$$u(t - t_0) \to y(t - t_0)$$
- If $y(t) = h(u(t))$, then
$$h(z(t - t_0)) = y(t - t_0)$$

Time-invariance proof example

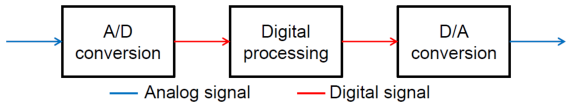
Time-invariant system

$$u(t) \xrightarrow{h} y(t) = u(t) + 7$$
$$h\big(u(t - t_0)\big) = u(t - t_0) + 7 = y(t - t_0)$$
$$u(t - t_0) \xrightarrow{h} y(t - t_0) \qquad \square$$

Time-variant system

$$u(t) \xrightarrow{h} y(t) = u(t) + t$$
$$h\big(u(t - t_0)\big) = u(t - t_0) + t$$
$$y(t - t_0) = u(t - t_0) + (t - t_0)$$
$$h\big(u(t - t_0)\big) \neq y(t - t_0) \qquad \square$$

Digitalization



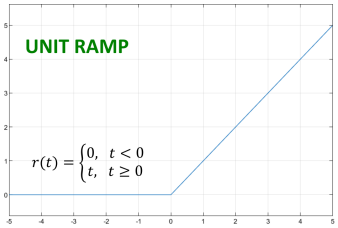
Time discretization

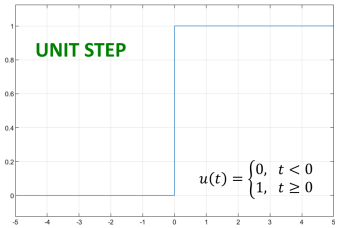
- Sampling: continuous signal measured at fixed time intervals
- Sampling rate: number of samples per second [Hz or kHz]

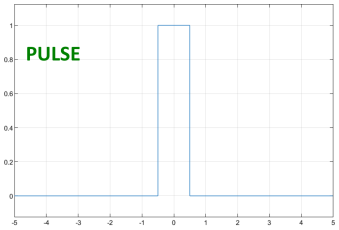
Amplitude discretization

- Quantization: signal values are rounded to discrete levels
- Quantization type: uniform uses equal steps; non-uniform uses variable steps
- Quantization step: coarse steps give lower precision; fine steps give higher precision

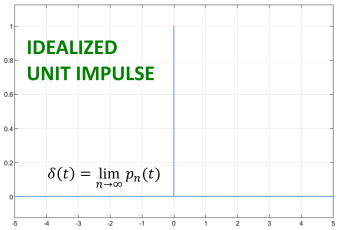
Basic input signals







$$p_n(t) = \begin{cases} \frac{n}{2}, & -\frac{1}{n} < t < +\frac{1}{n} \\ 0, & elsewhere \end{cases}$$



$$\delta(t) = \begin{cases} 1, & t = 0 \\ 0, & elsewhere \end{cases}$$

SW05: Linearization

Steps for linearizing a function

Step 1. Determine the nonlinear differential equation:
$$M\ddot{h} + c\dot{h} = Mg - kh^3$$

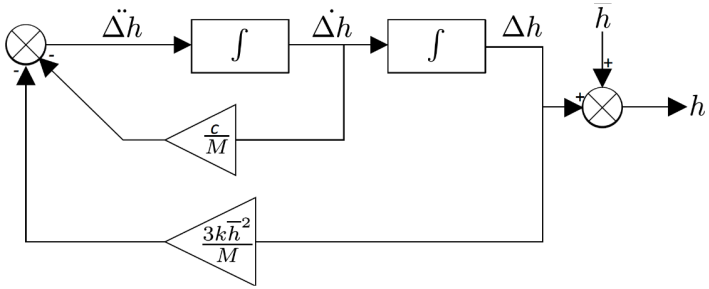
Step 2. Define $f(\ddot{h}, \dot{h}, h) = 0$:
$$M\ddot{h} + c\dot{h} + kh^3 - Mg = 0$$

Step 3. Define the slopes Δ:
$$\Delta\ddot{h} = \ddot{h} - \bar{\ddot{h}}$$

Step 4. Calculate the partial derivatives of f at the steady state for all the variables:
$$\frac{\partial f}{\partial \ddot{h}} = M\Delta\ddot{h}$$

Step 5. Establish a linear equation using the slopes solving for the highest differential order:
$$\Delta\ddot{h} = -\frac{c}{M}\Delta\dot{h} - \frac{3k(\bar{h})^2}{M}\Delta h$$

ODE diagram



SW06: Controllers

Analog controller

- Processes continuous signals using analog components, e.g. resistors, capacitors, and operational amplifiers
- Control signal: continuous voltage or current
- Main feature: fast real-time response with simple control logic

Digital controller

- Signal processing: uses A/D and D/A converters, causing time/amplitude discretization and delays
- Dynamic performance: slightly reduced due to discretization
- Flexibility: parameters can be updated by software
- Cost and size: smaller and often cheaper hardware
- Multitasking: can perform additional tasks besides control
- Noise robustness: digital filtering improves noise resistance
- Complexity: supports advanced control algorithms
- Maintenance: easier diagnostics and software updates
- Scalability: easy integration, upgrades, and expansion
- Precision: high accuracy and stable performance over time

Implementation platforms for digital controllers

- IPCs: industrial computers: complex real-time control tasks
- MCUs: compact, low-cost chips for simple embedded control
- PLCs: industrial controllers for automation in harsh env.
- DSPs: processors optimized for fast real-time signal calc.
- FPGAs: reconfigurable hardware for ultra-fast parallel control
- SBCs: compact computers: prototyping and low-cost control
- Cloud controllers: remote control, monitoring, and data analysis using cloud computing

System architecture and communication

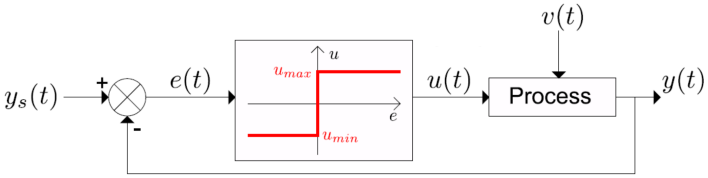
Centralized control

- One central unit handles signal processing, A/D and D/A conversion, inputs, and outputs
- Simple wiring, but possible bottlenecks in larger systems

Fieldbus communication

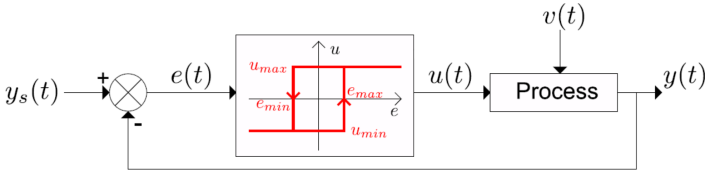
- Sensors, actuators, and controllers exchange data over an industrial network
- Scalable, modular, and reduces wiring complexity
- (e.g. ProfiNET, CAN/CANopen, EtherCAT)

2-point controller



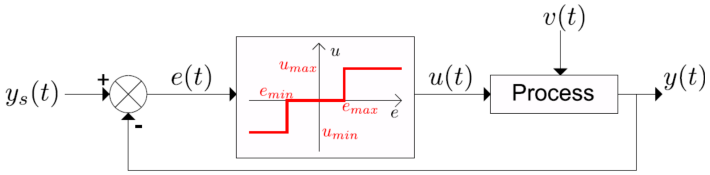
$$\begin{cases} u = u_{max} & \text{if } e > 0 \\ u = u_{min} & \text{if } e < 0 \end{cases}$$

2-point controller with hysteresis



$$\begin{cases} u = u_{max} & \text{if } e > e_{max} \\ u = u_{max} & \text{if } e_{min} < e < e_{max} \text{ and } \dot{e} < 0 \\ u = u_{min} & \text{if } e_{min} < e < e_{max} \text{ and } \dot{e} > 0 \\ u = u_{min} & \text{if } e < e_{min} \end{cases}$$

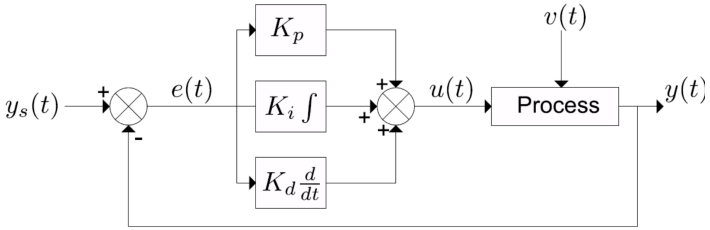
3-point controller



$$\begin{cases} u = u_{max} & \text{if } e > e_{max} \\ u = 0 & \text{if } e_{min} < e < e_{max} \\ u = u_{min} & \text{if } e < e_{min} \end{cases}$$

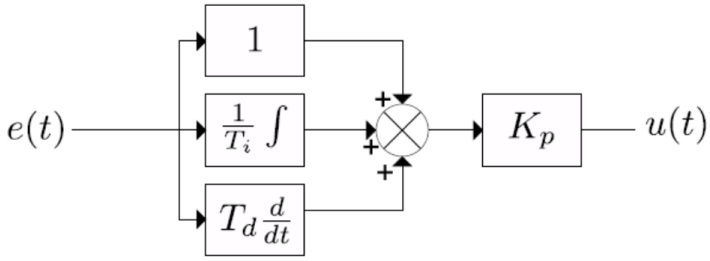
Proportional-Integral-Derivative (PID) controller

Mathematical parametrization



$$u(t) = K_p(e(t)) + K_i \int_0^t e(\tau)\, d\tau + K_d \frac{de(t)}{dt}$$

Technical usual parametrization



$$u(t) = K_p \left(e(t) + \frac{1}{T} \int_0^t e(\tau)\, d\tau + T_d \frac{de(t)}{dt} \right)$$

Controller gain: K_p

Integral time: $T_i = \frac{K_p}{K_i}$

Derivative time: $T_d = \frac{K_d}{K_p}$

PID principle

Proportional (P)

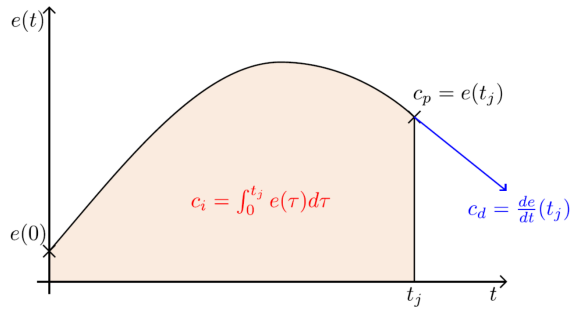
Larger error $e(t)$ gives a larger control output $u(t)$

Integral (I)

If $e(t) = 0$, the control output keeps increasing or decreasing to remove steady-state error

Derivative (D)

Larger error change rate $\dot{e}(t)$ gives a stronger control action to anticipate future error and reduce overshoot



$$u(t_j) = K_p c_p + K_i c_i + K_d c_d$$

$$u(t_j) = K_p \left(c_p + \frac{1}{T_i} c_i + T_d c_d \right)$$